TOPIC 2: MAIN COMPONENTS OF A MICROCOMPUTER

Official Undergraduate Lecture Slide Deck (Academic Year 2026/2027)

- Curriculum Coverage: Devices & Configurations (2.1) | Comparative Aspects (2.2)
- **Course Instructor: Madam Bertha Msuliche Lebalwa (MSc)**
- Assessment Structure: 2 Assignments (5%), 2 Tests (15%), Practical Report (10%), Practical Test (30%), UE (40%)*

**Assessment weightings are subject to adjustment by the course instructor.*

Topic 2: Lecture Outline & Learning Roadmap

Microcomputer Architecture, Subsystems & Comparisons (Coverage: 2.1, 2.2)

• Section 2.1: Microcomputer Devices & Logical Configurations

- Central Processing Unit (CPU): Control Unit, ALU, Internal Registers, Fetch-Decode-Execute
- Primary Memory: ROM (PROM, EPROM, Flash) vs RAM (SRAM, DRAM, DDR4/DDR5 architectures)
- System Bus Topology: Data, Address, and Control buses; Expansion buses (PCIe, SATA, USB)
- Input & Output Subsystems: Human-interface peripherals, agricultural transducers, displays, printers
- Secondary & Tertiary Storage: Magnetic HDDs, Solid-State Drives (NAND flash), and Optical media

• Section 2.2: Comparative Aspects of Computer Categories

- Microcomputers vs Enterprise Mainframes: Fault tolerance, RAS, and I/O channels
- Engineering Workstations vs Standard Microcomputers: ECC RAM, multi-socket CPUs, and pro GPUs
- Historical Minicomputers vs Modern Microcomputer Clusters and Cloud Nodes

TOPIC OBJECTIVE

Deconstruct the physical and logical anatomy of modern microcomputers, examining bus topologies, memory hierarchies, and storage physics.

Topic 2: Intended Learning Outcomes (ILOs)

Core Competencies for Engineering & Informatics Students

- Detail the internal micro-architecture of the CPU, including CU, ALU, and dedicated registers.
- Trace the step-by-step micro-operations of the Fetch-Decode-Execute instruction cycle.
- Evaluate the memory hierarchy in terms of access latency, capacity, and cost per bit.
- Calculate address bus capacity (2ⁿ) and theoretical bus bandwidth across PCIe and DDR standards.
- Contrast HDD mechanical physics (seek time, latency) with SSD semiconductor NAND flash architecture.
- Differentiate between Engineering Workstations, Consumer PCs, Mainframes, and Minicomputers.
- Integrate agricultural sensor interfaces (ADC, I2C, SPI) into microcomputer data collection pipelines.

Microcomputer Hardware Block Diagram

Logical and Physical Interconnects of a Modern Microcomputer System

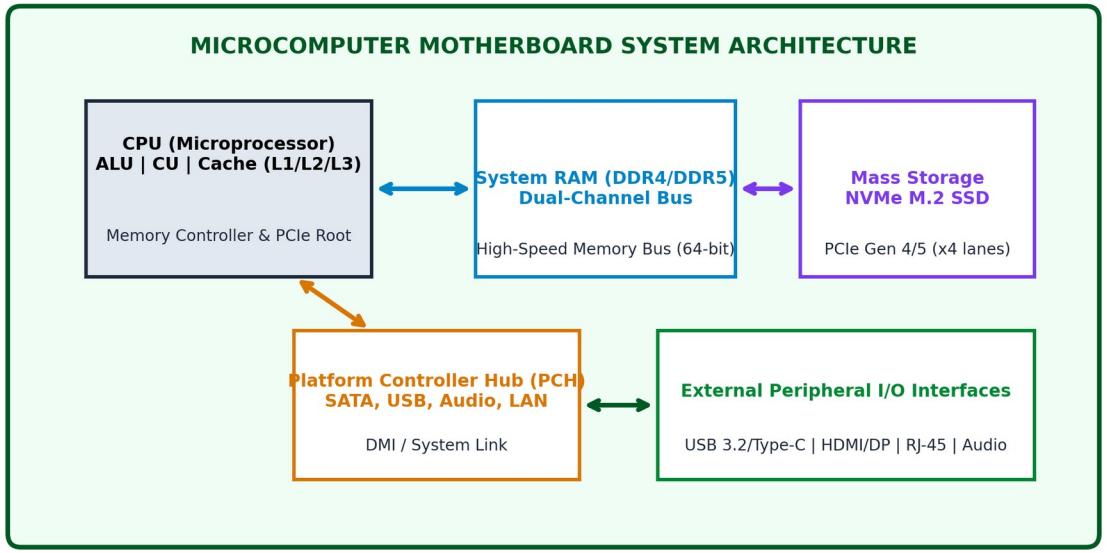


Figure: Functional block diagram showing CPU, System Buses, Memory, Storage, and I/O Subsystems.

ARCHITECTURAL ANALYSIS & PRINCIPLES

- Central Processing Unit (CPU): Executes instructions and orchestrates system timing.
- System Bus: The tri-bus interconnect (Address, Data, Control) linking all primary subsystems.
- Northbridge / System Agent: Connects high-speed devices (RAM, PCIe graphics) to the CPU.
- Southbridge / PCH (Platform Controller Hub): Connects slower I/O buses (SATA, USB, Audio, Ethernet, BIOS).

The Central Processing Unit (CPU): Core Subsystems

The Computational Engine of the Microcomputer

- Arithmetic Logic Unit (ALU): Executes binary arithmetic (+, -, *, /) and bitwise Boolean logic (AND, OR, XOR, NOT, shifts).
- Control Unit (CU): Decodes instructions, generates sequential micro-control signals, and manages clock synchrony.
- Internal Register File: High-speed, on-die storage cells operating at processor clock speed:
  - Program Counter (PC): Holds the memory address of the next instruction to fetch.
  - Memory Address Register (MAR): Holds the physical memory address currently being accessed.
  - Memory Data Register (MDR): Buffers data read from or written to memory.
  - Instruction Register (IR): Holds the operational code (opcode) currently being decoded.
  - Accumulator / General-Purpose Registers: Holds intermediate arithmetic operands and results.
  - Flags / Status Register: Indicates condition codes (Zero, Carry, Overflow, Sign, Interrupt enable).

The Instruction Cycle: Fetch, Decode, Execute

The Heartbeat of Microcomputer Processing

- 1. Fetch Phase:
  - `MAR ← [PC]`: Address in PC is placed onto the address bus.
  - `Memory Read`: Control unit asserts read signal; memory places instruction onto data bus.
  - `MDR ← Data Bus`: Instruction loaded into MDR.
  - `IR ← [MDR]`: Instruction transferred to Instruction Register.
  - `PC ← [PC] + 1`: Program counter incremented to point to the next instruction.
- 2. Decode Phase: Control unit decodes opcode in IR, determines required operands and addressing modes.
- 3. Execute Phase: ALU performs required operation; results written back to registers or main memory.
- 4. Interrupt Check: Processor verifies if external hardware interrupts (timer, I/O) are asserted.

The Computer Memory Hierarchy

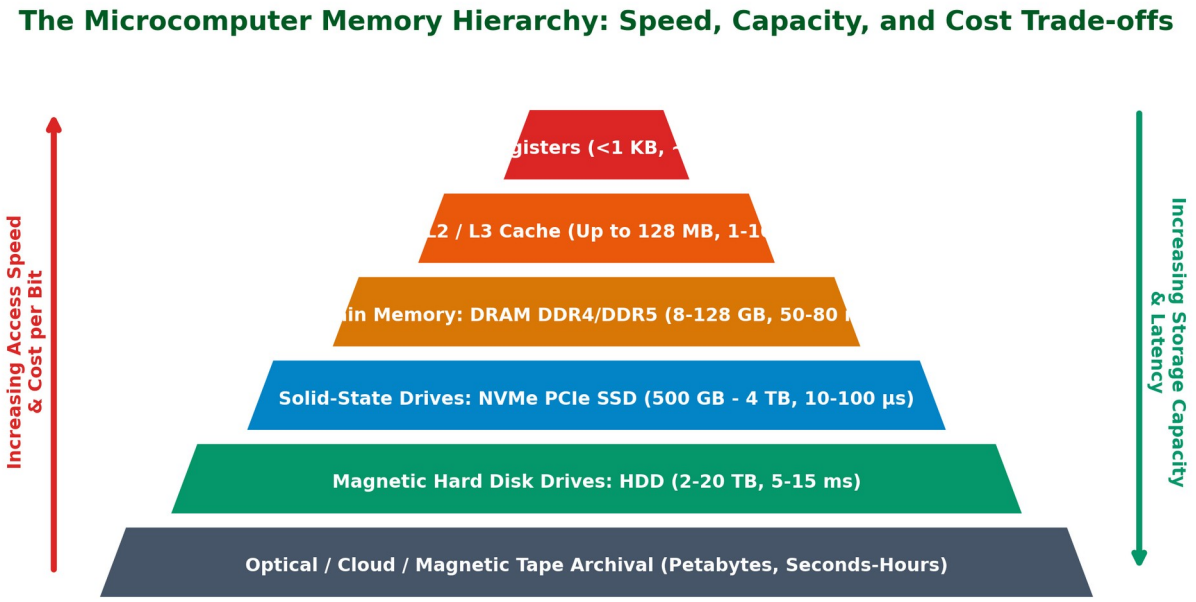


Figure: The Memory Hierarchy Pyramid balancing speed, capacity, and cost per bit.

ARCHITECTURAL ANALYSIS & PRINCIPLES

- Trade-off Principle: As you descend the pyramid, capacity increases, cost per bit decreases, but access latency increases exponentially.
- Level 0 - CPU Registers: < 1 KB, < 0.5 ns latency (direct silicon gates).
- Level 1 - L1 Cache: 32–64 KB per core, 1 ns latency (on-die SRAM).
- Level 2 - L2 Cache: 512 KB–1 MB per core, 3–5 ns latency.
- Level 3 - L3 Cache: 16–96 MB shared, 10–15 ns latency.
- Level 4 - Main Memory (DRAM): 16–128 GB, 50–70 ns latency.
- Level 5 - Non-Volatile Storage (NVMe SSD / HDD): 1–10 TB, 10 µs (SSD) to 10 ms (HDD).

Primary Memory: ROM Technologies

Non-Volatile Semiconductor Firmware Storage

- Read-Only Memory (ROM): Non-volatile primary storage retaining code even when power is removed.
- Evolution of ROM Devices:
  - Mask ROM: Programmed during silicon fabrication; permanently unalterable.
  - PROM (Programmable ROM): Field-programmable once using fusible links blown by high voltage.
  - EPROM (Erasable PROM): Erased by exposing quartz window to intense ultraviolet (UV) light for 20 minutes.
  - EEPROM (Electrically Erasable PROM): Byte-erasable electrically in-circuit; limited write cycles.
  - Flash ROM (NOR Flash): Block-erasable high-speed firmware storage powering modern UEFI BIOS chips.

Primary Memory: RAM Technologies (SRAM vs DRAM)

Static vs Dynamic Volatile Random-Access Memory

- Static RAM (SRAM):
  - Storage Element: 6-transistor (6T) flip-flop circuit.
  - Attributes: Fast (1–5 ns), does not require refresh cycles, low density, high power, very expensive.
  - Application: Exclusively utilized for on-die CPU cache hierarchies (L1, L2, L3).
- Dynamic RAM (DRAM):
  - Storage Element: 1 transistor and 1 tiny capacitor (1T-1C cell).
  - Attributes: High storage density, inexpensive, but capacitor charge leaks within milliseconds.
  - Refresh Circuitry: Requires periodic refresh cycles (every 64 ms) to restore charge, introducing latency.
  - Application: Main system memory (DIMM modules).

Modern Synchronous DRAM Evolution: DDR4 to DDR5

Double Data Rate Architecture and Architectural Enhancements

- Double Data Rate (DDR) Principle: Transfers data on both the rising and falling edges of the clock signal.
- DDR4 SDRAM:
  - Clock Speeds: 2133 to 3200 MT/s; Voltage: 1.2V.
  - Channel Architecture: Single 64-bit data bus per module.
- DDR5 SDRAM:
  - Clock Speeds: 4800 to 8400+ MT/s; Voltage: 1.1V.
  - Channel Architecture: Two independent 32-bit subchannels per DIMM, improving memory access concurrency.
  - On-Die ECC: Internal error correction on the DRAM die for enhanced cell reliability at extreme densities.
  - On-Module PMIC: Power Management Integrated Circuit moved from motherboard to the memory module.

System Bus Architecture: Tri-Bus Fundamentals

The Shared Highway for Data, Addresses, and Control

• Definition: A set of shared parallel or serialized electrical conductors transmitting signals between components.

• 1. Data Bus:

- Function: Carries actual data words and instruction operands between CPU, memory, and I/O devices.
- Directionality: Bidirectional (data flows to and from the CPU).
- Width: Typically 64 bits in modern microcomputers.

• 2. Address Bus:

- Function: Carries the physical memory address or I/O port identifier being selected.
- Directionality: Unidirectional (CPU asserts addresses; memory responds).
- Capacity Rule: An n-bit address bus can uniquely address 2ⁿ distinct memory locations.

• 3. Control Bus:

- Function: Transmits operational timing and status signals (Memory Read/Write, I/O Read/Write, Clock, Reset, Interrupt Requests).

Quantitative Address Bus & Bandwidth Calculations

Deriving Maximum Physical Address Space and Transfer Rates

- Address Space Derivation: Total Memory = 2ⁿ bytes (for byte-addressable memory):
  - 16-bit address bus (e.g., Intel 8080): 2¹⁶ = 65,536 bytes = 64 KiB.
  - 32-bit address bus (e.g., Intel 80386 / Pentium): 2³² = 4,294,967,296 bytes = 4 GiB.
  - 64-bit architecture (implementing 48 physical lines): 2⁴⁸ = 281,474,976,710,656 bytes = 256 TiB.
- Memory Bus Bandwidth Formula:
  - Bandwidth (GB/s) = (Bus Clock Frequency × Transfers per Cycle × Bus Width in Bits) / (8 bits/byte × 10⁹).
  - Dual-Channel DDR4-3200: 3,200 × 10⁶ transfers/s × 128 bits / 8 = 51.2 GB/s peak bandwidth.

Expansion Bus Architecture: PCI to PCI Express (PCIe)

The Paradigm Shift from Shared Parallel to Point-to-Point Serial

- Legacy PCI (Parallel): Shared parallel bus (32-bit, 33 MHz = 133 MB/s); suffered from clock skew across traces.
- PCI Express (PCIe) Architecture:
  - Point-to-Point Serial switched fabric connecting each device directly to the root complex.
  - Scalable Lane Architecture: Uses dedicated differential signaling pairs (x1, x4, x8, x16 lanes).
- PCIe Generation Bandwidths (per lane, each direction):
  - PCIe 3.0: 8.0 GT/s (~985 MB/s per lane; x16 slot = 15.75 GB/s).
  - PCIe 4.0: 16.0 GT/s (~1.97 GB/s per lane; x16 slot = 31.51 GB/s).
  - PCIe 5.0: 32.0 GT/s (~3.94 GB/s per lane; x16 slot = 63.02 GB/s).
- Applications: High-end graphics cards (x16), NVMe solid-state storage (x4), 10GbE network adapters (x4).

Secondary Storage: Magnetic Hard Disk Drives (HDD)

Electromechanical Platters, Cylinders, and Physics of Access

• Mechanical Architecture: Rigid rotating aluminum/glass platters coated with magnetic thin film, spinning on a central spindle (5400, 7200, 10000, 15000 RPM).

• Read/Write Heads: Aerodynamic flying heads floating nanometers above platters on an actuator arm.

• Formatting Geometry: Tracks (concentric rings), Sectors (standard 512 bytes or 4,096-byte Advanced Format), Cylinders (aligned tracks across all platters).

• Total Data Access Time Components:

- 1. Seek Time (T_{seek}): Time for actuator arm to move to target cylinder (avg 4–9 ms).
- 2. Rotational Latency (T_{rot}): Time for target sector to rotate under head (avg = 1/2 revolution; 4.17 ms at 7200 RPM).
- 3. Transfer Time (T_{trans}): Time to read/write sequential data bits (governed by track density).
- Average Access Time = T_{seek} + T_{rot} + Overhead ≈ 10 to 15 ms.

Secondary Storage: Solid-State Drives (SSD)

Semiconductor Flash Storage, NAND Physics, and Flash Controllers

• Core Architecture: Silicon NAND flash memory chips managed by an intelligent multi-core Flash Controller.

• Zero Moving Parts: Complete elimination of seek time and rotational latency; access time < 0.05 ms.

• NAND Flash Cell Technologies:

- SLC (Single-Level Cell): 1 bit/cell; highest endurance (100,000 P/E cycles), fastest, highest cost.
- MLC (Multi-Level Cell): 2 bits/cell; 3,000–10,000 P/E cycles.
- TLC (Triple-Level Cell): 3 bits/cell; 1,000–3,000 P/E cycles; dominant consumer standard.
- QLC (Quad-Level Cell): 4 bits/cell; ~1,000 P/E cycles; high capacity, lower endurance.

• Flash Management Imperatives: Wear-Leveling (distributing writes evenly across cells) and TRIM command (notifying drive of deleted LBAs to enable background block garbage collection).

Comparative Analysis: Storage Interfaces & Technologies

Metric	Magnetic HDD (SATA)	SATA SSD (2.5")	NVMe SSD (PCIe 4.0)	Magnetic Tape (LTO-9)
Storage Medium	Magnetic Platters	NAND Flash	3D NAND Flash	Magnetic Barium Ferrite Tape
Sequential Read	150 - 260 MB/s	500 - 560 MB/s	5,000 - 7,400 MB/s	400 MB/s (native)
Random IOPS	75 - 150 IOPS	50,000 - 90,000 IOPS	500,000 - 1,000,000 IOPS	N/A (Sequential Only)
Access Latency	10 - 15 ms	50 - 100 µs	10 - 20 µs	Seconds to Minutes
Vulnerability	Mechanical Shock	Cell Wear (P/E)	Thermal Throttling	Physical Misplacement
Primary Role	Bulk Secondary / NAS	Budget Upgrades	OS Boot & Pro Compute	Cold Deep Archival

Input Subsystems: Transducers to Digital Interface

Human Input Devices & Agricultural Sensors

• Keyboards: Switch matrix scanned by dedicated microcontroller; debounce circuitry generates make/break scan codes.

• Optical Mice: LED/laser illuminates surface; CMOS optical sensor captures thousands of surface images/sec, calculating relative movement vectors.

• Agricultural Sensors & Transducers:

- Thermistors & RTDs: Temperature-dependent resistance measuring grain silo thermal gradients.
- Capacitive Soil Moisture Probes: Measures dielectric permittivity of soil water content.

• Analog-to-Digital Converter (ADC): Quantizes continuous analog voltages (0–5V) into discrete binary words (e.g., 10-bit ADC provides 1,024 discrete levels).

• Communication Protocols: I2C (Inter-Integrated Circuit - 2 wire), SPI (Serial Peripheral Interface - 4 wire), UART (Asynchronous Serial).

Output Subsystems: Displays and Printing Technologies

Visual Telemetry, Rasterization, and Hardcopy Reproduction

- Display Technologies:
  - LCD (Liquid Crystal Display) with LED Backlight: In-Plane Switching (IPS) panels offering superior color accuracy and 178° viewing angles for GIS/CAD.
  - OLED (Organic LED): Self-emissive pixels offering infinite contrast ratio, true blacks, and rapid response times.
  - Key Metrics: Native resolution (Full HD 1080p, 4K UHD), aspect ratio (16:9, 16:10), refresh rate (60 Hz, 144 Hz).
- Printing Technologies:
  - Dot-Matrix (Impact): Uses print pins striking an ink ribbon; indispensable for multi-part carbon copy financial invoices at SUA bursar.
  - Inkjet: Piezoelectric or thermal nozzles spraying microscopic picoliter droplets; high-resolution photographic rendering.
  - Laser (Electrophotographic): Laser discharges photosensitive drum, attracting toner powder transferred and fused to paper via heated rollers.

Microcomputers vs Enterprise Mainframes

Contrasting Architectural Paradigms and Throughput Philosophy

• Processing Philosophy:

- Microcomputer: Optimized for low-latency interactive processing, single-user responsiveness, and high single-thread clock speed.
- Mainframe: Engineered for massive concurrent throughput, processing billions of independent transactional records per second.

• I/O Channel Offloading:

- Microcomputer: CPU actively participates in managing bus interrupts and storage transfers (DMA helps, but CPU coordinates).
- Mainframe: Employs hundreds of dedicated Channel Processors (System Assist Processors) that handle all I/O independently from main compute engines.

• Redundancy & Fault Tolerance: Mainframes feature dual lockstep execution, redundant power/cooling, and dynamic processor sparing.

Engineering Workstations vs Standard Microcomputers

Mission-Critical Precision Computing versus Consumer Utility

• Processor Architecture: Consumer PCs utilize Intel Core i5/i7 or AMD Ryzen; Workstations deploy Intel Xeon or AMD Threadripper with higher PCIe lane counts.

• Memory Integrity (ECC RAM):

- Standard PCs: Non-ECC RAM; silent bit-flips caused by cosmic radiation can corrupt simulation data or cause system crashes.
- Workstations: Error-Correcting Code (ECC) memory detects and corrects single-bit errors automatically, halting on multi-bit errors.

• Graphics Accelerators: Consumer GPUs (Nvidia GeForce) prioritize gaming frame rates; Professional GPUs (Nvidia RTX A-Series) feature ISV-certified drivers for AutoCAD, ArcGIS, and SolidWorks with double-precision floating-point accuracy.

• Mean Time Between Failures (MTBF): Workstation components are binned and rated for continuous 24/7/365 full-load computational workloads.

Minicomputers vs Modern Microcomputer Clusters

The Evolution from Midrange Hardware to Distributed Micro-Nodes

• The Minicomputer Era (1965–1985): Multi-user systems (DEC PDP-11, VAX) sharing compute power via serial 'dumb terminals' across university departments.

• The Downsizing Revolution: Microprocessors became so powerful and inexpensive that client-server microcomputer networks replaced monolithic minicomputers.

• Modern Microcomputer Clusters (Beowulf Clusters):

- Network of commodity microcomputer nodes interconnected via high-speed gigabit switches running Linux.
- Delivers supercomputing throughput at a fraction of minicomputer or mainframe acquisition costs.
- Applied extensively in agricultural genomics, crop disease prediction, and remote sensing imagery processing at SUA.

Worked Engineering Example: HDD Access Time vs SSD

Mathematical Derivation of Secondary Storage Latency

- Scenario: Read 1,000 randomly scattered 4 KB files from disk.
- Magnetic HDD Specifications: 7,200 RPM, Average Seek Time = 8.5 ms, Transfer Rate = 150 MB/s.
  - Rotational Time per rev = 60 / 7200 = 8.33 ms. Average Rotational Latency = 8.33 / 2 = 4.17 ms.
  - Transfer Time per 4 KB = 0.004 MB / 150 MB/s = 0.027 ms.
  - Total Time per File = 8.5 + 4.17 + 0.027 ≈ 12.7 ms.
  - Total Time for 1,000 Random Files = 1,000 × 12.7 ms = 12.7 seconds.
- NVMe SSD Specifications: Random Read Latency = 0.03 ms.
  - Total Time for 1,000 Random Files = 1,000 × 0.03 ms = 0.030 seconds (30 milliseconds).
- Result: The SSD is over 420 times faster for random small-file database operations!

Agricultural IoT Hardware Design Case Study

Autonomous Soil & Microclimate Station at SUA Mazimbu Farm

• Environmental Challenge: Tropical field deployment, high humidity, erratic cellular coverage, no grid AC power.

• System Architecture:

- Embedded Microcomputer: ESP32 dual-core microcontroller (240 MHz, ultra-low-power deep sleep mode).
- Sensor Bus (I2C): SHT31 air temperature/humidity sensor + BH1750 solar lux transducer.
- Analog Sensor Subsystem: Capacitive soil moisture probes interfaced via ADS1115 16-bit external ADC.
- Power Subsystem: 20W Monocrystalline solar panel + MPPT charge controller + 18650 LiFePO4 battery pack.
- Telemetry Subsystem: LoRaWAN 868 MHz transmitter beaming packets 5 km to the DIIT gateway receiver.

Instruction Pipeline Architecture & Pipeline Hazards

Overlapping Instruction Phases to Maximize Throughput

• Pipelining Principle: Divides instruction execution into sequential stages (Fetch, Decode, Execute, Memory, Write-back), allowing multiple instructions to execute concurrently across different stages.

• Pipeline Hazards (Factors causing pipeline stalls / bubbles):

- 1. Structural Hazard: Hardware resource conflict (e.g., single memory port simultaneously accessed by Fetch and Memory stages).

- 2. Data Hazard (Read-After-Write - RAW): Instruction depends on the computational result of an uncompleted prior instruction; mitigated via data forwarding/bypassing.

- 3. Control Hazard (Branch Hazard): Pipelining branch instructions before condition outcome is evaluated; mitigated via Branch Prediction (static vs 2-bit dynamic branch history tables) and Speculative Execution.

Microprocessor Packaging & Socket Standards

LGA, PGA, and BGA Thermal and Interconnect Engineering

• Pin Grid Array (PGA): Pins located on the bottom of the processor package that mate with socket holes (e.g., legacy AMD AM4); pins fragile and vulnerable to bending.

• Land Grid Array (LGA): Pins located inside the motherboard socket; processor package features flat gold contact pads (e.g., Intel LGA1700, AMD AM5); reduces risk of damaged CPU pins.

• Ball Grid Array (BGA): Processor permanently soldered to motherboard using microscopic solder balls; ubiquitous in ultra-thin laptops, tablets, and smartphones; non-upgradeable.

• Thermal Interface Materials (TIM) & Heat Spreaders: Integrated Heat Spreader (IHS), thermal paste, and finned heatsinks with copper heatpipes dissipating heat to prevent thermal throttling.

Storage Protocols: SATA AHCI vs NVMe Architecture

From Spinning Disk Legacy to Silicon-Optimized Queuing

- Advanced Host Controller Interface (AHCI):
  - Designed in the 1990s specifically for high-latency rotating magnetic hard drives.
  - Bottlenecks: Single command queue with maximum depth of 32 commands; high software interrupt overhead (over 20,000 CPU cycles per I/O).
- Non-Volatile Memory Express (NVMe):
  - Architected from the ground up for low-latency solid-state NAND flash directly over PCIe lanes.
  - Unrivaled Scalability: Up to 64,000 independent command queues, each capable of holding 64,000 commands.
  - Multi-Core Parallelism: Each CPU core gets its own dedicated submission and completion queue pair, completely eliminating lock contention.

Peripheral Bus Evolution: USB 2.0 to USB4 & Thunderbolt

The Universal Interconnect: Bandwidth Scaling and Power Delivery

- USB 2.0 (HighSpeed, 2000): 480 Mbps half-duplex; 4-wire interface; maximum 2.5W power delivery (5V @ 500mA).
- USB 3.2 Gen 1 (SuperSpeed): 5 Gbps full-duplex with dedicated transmitter and receiver differential pairs.
- USB 3.2 Gen 2x2: 20 Gbps using dual-lane Type-C cabling.
- USB4 & Thunderbolt 4 / 5 Architecture:
  - 40 Gbps to 80 Gbps bidirectional bandwidth.
  - Protocol Tunneling: Simultaneously tunnels PCIe data, DisplayPort 2.1 video streams, and USB packets over a single reversible Type-C cable.
  - USB Power Delivery (USB-PD 3.1): Up to 240 Watts (48V @ 5A), powering high-performance laptops and charging field agricultural equipment.

Display Standards: DisplayPort vs HDMI vs Analog VGA

Digital Packetized Video Streams and Color Depth

- Analog Legacy: VGA (15-pin D-sub) transmitting continuous analog RGB voltages; prone to signal attenuation and EMI ghosting.
- HDMI (High-Definition Multimedia Interface):
  - Consumer audio/video standard; HDMI 2.1 supports up to 48 Gbps (Fixed Rate Link), enabling 4K @ 120 Hz and 8K @ 60 Hz.
- DisplayPort (VESA DP 2.1):
  - Professional computing standard; packetized micro-packet architecture (similar to PCIe and Ethernet); DP 2.1 supports up to 80 Gbps (UHBR 20).
  - Multi-Stream Transport (MST): Allows daisy-chaining multiple independent monitors from a single physical port.
- Color Precision: 8-bit color (16.7 million colors) vs 10-bit HDR (1.07 billion colors) for precision GIS satellite analysis.

Motherboard Form Factors & Power Delivery (VRM)

Chassis Geometry and Clean Multi-Phase DC Power

• Motherboard Form Factors:

- ATX (Standard): 305 × 244 mm; up to 7 expansion slots, 4 RAM DIMMs; used in departmental servers and workstations.
- Micro-ATX (mATX): 244 × 244 mm; 4 expansion slots; cost-effective balance for computer lab desktops.
- Mini-ITX: 170 × 170 mm; 1 expansion slot; used in compact industrial PCs and automated tractor cab telemetry units.

• Voltage Regulator Module (VRM):

- Step-down buck converter stepping down 12V DC from ATX power supply to ~1.0–1.3V required by modern CPU cores.
- Multi-Phase Architecture: Uses multiple parallel phases (PWM controller, MOSFETs, chokes, solid capacitors) to deliver clean, ripple-free current exceeding 150 Amps without overheating.

Hardware Accelerators: NPUs & Systolic Arrays

Dedicated Silicon for On-Device Machine Learning and AI

• Neural Processing Unit (NPU) / Tensor Processing Unit (TPU):

  - Specialized silicon execution blocks designed specifically for low-precision tensor operations (INT8, FP16, BF16).

• Systolic Array Architecture: 2D grid of processing elements where data flows synchronously through cells, multiplying and accumulating without constantly reading/writing back to main memory registers.

• Extreme Energy Efficiency: Delivers 10 to 45 TOPS (Trillions of Operations Per Second) at just 3 to 10 Watts.

• Agricultural Application: Real-time computer vision on battery-powered edge drones at SUA, classifying crop disease lesions and weed species instantly in the field without cloud connectivity.

Firmware Diagnostics: POST, Beep Codes & Diagnostic LEDs

Troubleshooting Microcomputer Hardware Boot Failures

• Power-On Self-Test (POST): Sequence of firmware diagnostic routines executed immediately upon applying electrical power.

• Subsystem Verification Sequence:

1. CPU registers and ALU sanity check.
2. BIOS ROM checksum verification.
3. System timer and DMA controller initialization.
4. Primary DRAM detection and read/write parity verification.
5. Video display controller initialization and VRAM detection.

• Diagnostic Signaling:

- Beep Codes: Internal piezo speaker signals failure before video is initialized (e.g., 1 long 2 short = video card failure).
- Motherboard Q-LEDs: Discrete LEDs (CPU, DRAM, VGA, BOOT) indicating which subsystem halted the boot process.
- Seven-Segment POST Code Displays: Displays hexadecimal status codes (e.g., code `00` = CPU failure; code `55` = Memory not installed).

Topic 2: Key Takeaways & Synthesis Matrix

Essential Hardware Principles for Technical Mastery

- CPU Architecture: CU orchestrates; ALU computes; registers provide ultra-fast operand buffering.
- Instruction Pipeline: Fetch (PC to MAR, memory to MDR, MDR to IR, PC increment) → Decode → Execute.
- Memory Hierarchy: L1/L2/L3 SRAM Cache (fastest, costly) → DRAM System Memory → NVMe/SATA SSD → Magnetic HDD.
- Bus Physics: Address Bus determines maximum memory capacity (2ⁿ); Data Bus width determines word size.
- Workstation Differentiators: ECC memory prevents silent corruption; ISV workstation GPUs guarantee geometric precision.
- Storage Mechanics: HDDs bounded by mechanical physics (seek + rotation); SSDs bounded by NAND flash controllers and bus protocol.

Review Quiz & Knowledge Check

Self-Testing Key Architectural Principles

1. If a microprocessor has a 36-bit address bus, what is the maximum physical RAM it can directly address? (Ans: 64 GiB).
2. Why does Dynamic RAM require periodic electrical refresh cycles, whereas Static RAM does not?
3. Explain the function of the Program Counter (PC) during the instruction fetch phase.
4. What happens when an NVMe SSD experiences severe thermal throttling during sustained sequential writes?
5. Why is ECC memory mandatory for scientific GIS servers but considered unnecessary for consumer gaming laptops?

Reading Assignments & Homework Notice

Preparation for Next Lecture & Continuous Assessment

- Required Readings:
  - Stallings, W. - Computer Organization and Architecture (11th Ed.), Chapters 3, 4, 5 & 7.
  - Hennessy & Patterson - Computer Architecture: A Quantitative Approach, Chapter 2 (Memory Hierarchy).
- Homework Assignment 2 Notice:
  - Available on SUA Moodle / Assignment directory: `BIT_111_Assignment_2.pdf`.
  - Total Allocation: 100 Marks (Covers Subsystem Calculations, Storage Latency, and Workstation Sizing).
  - Submission Deadline: End of Week 4 by 17:00 EAT.
- Next Lecture Preview: Topic 3 — Personal Computer (PC) Security Controls (Firmware, TPM, BitLocker, Access Models).